IAS Physics

Topic 2: Materials

2A: Fluids (1. Fluids, Density, and Upthrust)

23) **Density** is the volumic mass of an object, depending on how closely the particles within a substance are packed and the mass of those particles.

Density (**ρ**) = Mass (**m**) / Volume (**V**)

ρ (kg/m^3) = **m** (kg) / **V** (m^3)

24) **Upthrust** is the upward force acting on a submerged object, due to fluid pressure. The force of upthrust is equal to the weight of fluid displaced by the object that has been submerged.

Archimedes' principle states that, when an object is completely submerged, the **mass** of fluid displaced is equal to the volume of an object multiplied by the fluid's density.

Mass of Fluid Displaced (**m**) = Density of fluid (**ρ**) x Volume of Submerged Object (**V**)

We can therefore find the **weight** of the fluid displaced (**upthrust**) by multiplying our value for mass with **g**.

Upthrust (**U**) = Density of fluid (**ρ**) x Volume of Submerged Object (**V**) x 9.81 (**g**)

U (N) = **ρ** (kg/m^3) x **V** (m^3) x **g**

2A: Fluids (4. Terminal Velocity)

25)

a) To find the **viscous drag** acting on a small sphere at low speeds, **Stoke's law** is used:

Viscous Drag (***F***) = 6π x Coefficient of Fluid Viscosity (η) x Radius of Sphere x Velocity of Sphere (**v**)

$$\boldsymbol{F}\,(\mathrm{N}) = \mathbf{6} \times \boldsymbol{\pi} \times \boldsymbol{\eta}\,(\mathrm{Pa\ s}) \times \mathbf{r}\,(\mathrm{m}) \times \mathbf{V}\,(\mathrm{m/s})$$

b) This equation only applies to <u>small, spherical objects moving at low speeds with laminar flow</u> (in the absence of turbulent flow).

Viscosity is also temperature dependent. For gases, viscosity increases with temperature.
For liquids, viscosity is inversely proportional to temperature. Liquids have a decrease in viscosity as temperature increases (think of warming oil in a skillet).

26) **Core Practical 2:** Using a Falling-Ball Method to Determine the Viscosity of a Liquid



Method:

1. Weigh each ball and measure their radii (**r**), then use that value to calculate their densities (**ρ**) by dividing mass by volume.

2. Place three rubber bands around a tube. The highest must be below the liquid's surface, so that the ball will be at terminal velocity. Position the other two bands should be spaced apart so that two times can be measured. This will allow two values for terminal velocity to be calculated, so we can be sure the ball has reached terminal velocity.

3. Measure the distance between the rubber bands. d_1 will be the distance between the highest band and the middle one, and d_2 will be the distance between the highest band and the lowest one.

4. Release a ball into the liquid, starting the timer when the ball passes the highest rubber band. t_1 will be the time for it to reach the middle band,

and t_2 will be the time taken for it to reach the lowest band.

5. Repeat **step 4** at least three times for each ball.

6. Calculate the mean values for t_1 and t_2 for each ball, then divide d_1 and d_2 by these values to find the mean terminal velocities for each ball.

7. **Viscosity** can then be calculated using the relationships between drag, weight, and upthrust at terminal velocity.

SAFETY:

- Have paper towels for cleaning up liquid spills to avoid slipping.
- Secure the tube to avoid it falling over.
- Avoid skin contact with mineral or motor oils, if used.

At terminal velocity for a falling ball, weight = upthrust + drag.
The weight of each ball can simply be calculated by multiplying its mass with **g**, and upthrust is calculated as **ρVg**.

Using the formula for drag, **6πηrV**, the only unknown we will have is **η**, its **viscosity**.

By re-arranging our equation to be W - U = D, we can easily find **viscosity** by dividing W-U by **6πrV**.

2B: Solid Material Properties (1. Hooke's Law)

27) **Hooke's Law** states that the force needed to extend a spring is proportional to the extension of the spring. This law applies until the limit of proportionality.

An object that has undergone **elastic deformation** will return to its original size and shape once the force is removed, as long as the **elastic limit** has not been passed.

Force Applied (Δ**F**) = Stiffness Constant (**k**) x Extension (Δ**x**)

ΔF (N) = **k** (N/m) x **ΔX** (m)

2B: Solid Material Properties (2. Stress, Strain, and the Young Modulus)

28) (Tensile or compressive) **Stress** measures the force within the cross-sectional area of a material sample, so that comparisons can be made between samples that are different sizes.

Stress (**σ**) = Force (**F**) / Cross-Sectional Area (**A**)

σ (Pa **OR** N/m^2) = **F** (N) ÷ **A** (m^2)

(Tensile or compressive) **Strain** measures the extension or compression of a material sample, taking its original length into account.

Strain (**ε**) = Extension (**Δx**) ÷ Original Length (**x**)

ε (No unit) = **Δx** (m) ÷ **x** (m)

Young modulus measures the **stiffness** of a material. Stiffness relates to how much a material will deform when a force is applied to it.

Young Modulus (**E**) = Stress (**σ**) ÷ Strain (**ε**)

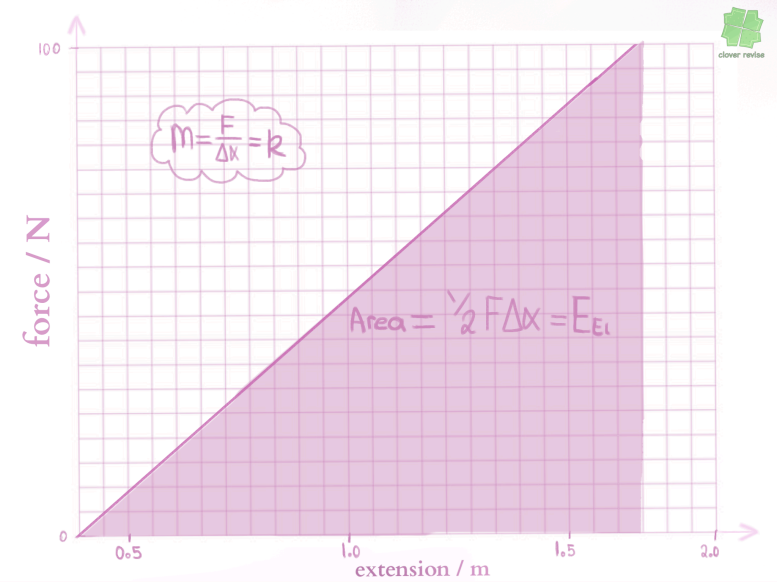
E (Pa) = **σ** (Pa **OR** N/m²) ÷ **ε** (No unit)

By substituting our previous equations into the equation for Young modulus, we see that Young modulus can also be given as:

E (Pa) = (**F** (N) x **x** (m)) ÷ (**A** (m) x **Δx** (m))

2B: Solid Material Properties (1. Hooke's Law)

29)

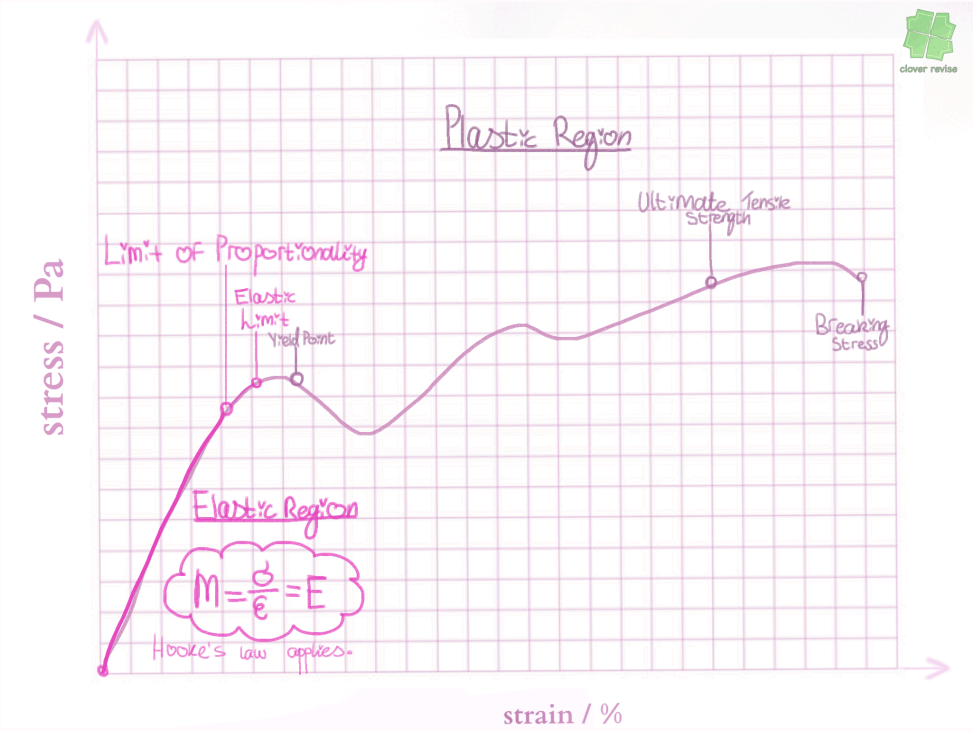


The gradient of a force-extension / force-compression graph is the stiffness constant (**k**).

The area (½ **FΔx**) beneath the graph gives the work done, or the **elastic strain energy**. For a non-linear graph, this is found by counting squares.

2B: Solid Material Properties (3. Stress-Strain Graphs)

30)



The **limit of proportionality** is the final point wherein Hooke's law applies. Although the material may still have elastic behaviour, it may not be proportional to the stress or force (no longer linear).

The **elastic limit** is the point directly before the material is permanently deformed. After this point, even when stress or force is released, the material will not return to its original size nor shape.

The **yield point** is the point wherein a material will have a large increase in extension for a relatively smaller force, due to a re-organisation of its atomic substructure.

The **Ultimate Tensile Stress** (**UTS**) is the highest possible stress within a material.

The **breaking stress** or fracture stress is the stress within a material when it breaks.

For a stress-strain graph, the <u>gradient</u> is the **Young Modulus** of the material.

31) **CORE PRACTICAL 3**: Determine the Young Modulus of a Material



Method:

1. Measure the diameter of the wire, *d*.

2. Attach the bench pulley to the end of a bench. Secure one end of your wire between two wooden blocks that have been fixed to the bench, around <u>three metres away from the pulley</u>.

3. Lay the wire out so it passes over the pulley, then attach the slotted mass hanger to the end.

4. Lay the metre rule near the pulley, under the wire, and attach a sticky label to mark the length. The length should be judged by looking vertically down, over the label's edge, to scale with the metre rule. A set square is used to avoid parallax errors by ensuring a straight distance is being measured.

5. From the wooden blocks to the label's edge, measure the length of the wire, L.

6. Add a mass to the hanger, then record the difference between the label and the metre rule to calculate extension, Δx.

7. Repeat **step 4**, adding one mass at a time and recording the extension for each mass.

SAFETY:

- Wear eye protection and try not to overload the wire to avoid it causing injury by snapping.
- Do not exceed the maximum load.
- Have a box filled with bubble wrap or crumpled paper below the hanging masses to avoid foot injuries.

Plotting a graph of mass against extension will aid in calculating the **Young Modulus**, only using the linear portion of the graph.

The cross-sectional **area** of the wire can be calculated by using $\pi(\frac{1}{2}\,d)^2$, and the **force** can be found by multiplying mass by **g**. With force and area, **stress** can then be calculated for two separate points on the graph.

Two values for **strain** can be calculated by dividing the extension at two different points by the wire's original length.

Finally, **Young Modulus** can be calculated by dividing the difference in stress values by the difference in strain values ($\mathbf{E} = \Delta\boldsymbol{\sigma} \div \Delta\boldsymbol{\varepsilon}$).

2B: Solid Material Properties (1. Hooke's Law)

32) **Elastic strain energy** is energy stored within a material, from the work done to deform it, before it reaches its elastic limit.

$$\text{Elastic Strain Energy } (\Delta E_{EI}) = \tfrac{1}{2} \times F(\mathbf{N}) \times \text{Extension } (\Delta \mathbf{x})$$

$$\mathbf{E_{EI}}\ (\text{J}) = \tfrac{1}{2} \times \mathbf{F}\ (\text{N}) \times \Delta\mathbf{x}\ (\text{m})$$

The work done can also be found by calculating the area under a force-extension graph.